26/3/19 :-

Best Unbiased Estimation problem –

$X \sim P_\theta$, $\theta \in \Theta \subseteq \mathbb{R}$

$W : X \longrightarrow W(X)$

ϕ W is unbiased if $\forall \theta$ $E_\theta(W(X)) = \theta$

and W is best unbiased if :

(i) W unbiased

(ii) $\forall \theta$, $Var_\theta W(X) \leq Var_\theta W'(X)$ $\forall$ unbiased W'.

Theorem: CRLB :-

Suppose $X \sim P_\theta$, $\theta \in \Theta \subseteq \mathbb{R}$, and let $W(X)$ be an estimator of $g(\theta)$, satisfying:

(i) $\forall \theta$, (a) $\frac{d}{d\theta} E_\theta[W(X)] = \int_{\mathbb{R}} \frac{\partial}{\partial\theta}\left(W(x) P_\theta(x)\right) dx$

(b) $0 = \int_{\mathbb{R}} \frac{\partial}{\partial\theta}\left(P_\theta(x)\right) dx$

(ii) $\forall \theta$, $Var_\theta W(X) < \infty$

Then, $\forall \theta$ $Var_\theta[W(X)] \geq \dfrac{\left(\frac{d}{d\theta} E_\theta(W(X))\right)^2}{E_\theta\left[\frac{\partial}{\partial\theta}\log P_\theta(X)\right]^2}$

→ In particular, if $W(X)$ is an unbiased estimation of θ, then $\forall\theta$ $E_\theta W(X) = \theta$

and hence, CRLB ⇒ $Var_\theta W(X) \geq \dfrac{1}{E_\theta\left(\frac{\partial}{\partial\theta}\log P_\theta(X)\right)^2}$

Proof: Cauchy - ~~[illegible]~~ schwarz inequality :-

→ for two random variables A, B on the same prob space

$E(AB)^2 \leq E[A^2] \cdot E[B^2]$

$\left(\sum_i a_i b_i\right)^2 \leq \left(\sum_i a_i^2\right)\left(\sum_j b_j^2\right)$

$\left\{ \text{proof: } \forall \alpha \in \mathbb{R} : E\left[(A-\alpha B)^2\right] \geq 0 \right\}$

$\Rightarrow$ $\forall$ random variables $\tilde{X}, \tilde{Y}$

$$\operatorname{Cov}(\tilde{X},\tilde{Y})^2 \leq \operatorname{Var}(\tilde{X}) \cdot \operatorname{Var}(\tilde{Y})$$

$$\Rightarrow \quad \operatorname{Var}(\tilde{X}) \geq \frac{\operatorname{Cov}(\tilde{X},\tilde{Y})^2}{\operatorname{Var}(\tilde{Y})}$$

apply this with $\tilde{X} : w(x)$ $\quad$ $X \sim P_\theta$

$\tilde{Y} : \frac{\partial}{\partial\theta} \log P_\theta(x)$ $\quad$ $w(x)$ – RV

LHS : $\operatorname{Var}(\tilde{X}) = \operatorname{Var}_\theta\left(w(x)\right)$

RHS : $\operatorname{Cov}(\tilde{X},\tilde{Y}) = \operatorname{Cov}_\theta\left(w(x), \frac{\partial}{\partial\theta} \log P_\theta(x) \right)$

$$E_\theta(\tilde{Y}) = \int \frac{\partial}{\partial\theta} \log P_\theta(x) \cdot P_\theta(x)\, dx$$

$$= \int \frac{1}{\cancel{P_\theta(x)}} \frac{\partial}{\partial\theta} P_\theta(x) \cdot \cancel{P_\theta(x)}\, dx$$

$$= 0$$

by assumption 1b, i.e mean of RV $\tilde{Y}$ is zero

note: $\operatorname{Cov}(P,Q) = E\left[P_{\tilde{x}}(Q - E(Q))\right]$

i.e no need to centre the both.

$$= E_\theta\left[w(x) \cdot \frac{\partial}{\partial\theta} \log P_\theta(x) \right]$$

$$= \int w(x) \frac{1}{\cancel{P_\theta(x)}} \frac{\partial}{\partial\theta} P_\theta(x) \cdot \cancel{P_\theta(x)}\, dx$$

$$= \int \frac{\partial}{\partial \theta} \left(w(x) \, P_\theta(x) \right) dx$$

$$= \frac{d}{d\theta} E_\theta \left[w(x) \right]$$

by assumption 1.a

also $\quad E[\tilde{Y}^2] = \text{Cov}(\tilde{Y}) \quad \because$ Zero mean

$$= E_\theta \left(\frac{\partial}{\partial \theta} \log P_\theta(x) \right)^2$$

Hence proved

Remarks :-

1) for vector estimation problems

the CRLB is —

let $\underline{\theta} \in \Theta \subseteq \mathbb{R}^d$

$X \sim P_{\underline{\theta}}$, $w(X)$ an estimator with

(i) $$\frac{\partial}{\partial \theta_i} E_{\underline{\theta}}\left[h(x) \right] = \int \frac{\partial}{\partial \theta_i} \left[h(x) \, P_{\underline{\theta}}(x) \right] dx$$

$\forall \quad 1 \leq i \leq d$ and

for $h \equiv w_i \, / \, h \equiv 1$

(ii) $$\text{Var}_{\underline{\theta}} \, w_i(x) \leq \infty \quad \forall \, i, \underline{\theta}$$

$$\cancel{\therefore \neq \sum_{i=1}^{d} \text{Var}_\theta (w_i(x))}$$

Then, $$\underbrace{\text{Var}_{\underline{\theta}} \left[w(x) \right]}_{:= \sum_{i=1}^{d} \text{Var}_\theta (w_i(x))} \geq \left(\nabla_{\underline{\theta}} \psi \right)^T \cdot \left(I(\underline{\theta}) \right)^{-1} \cdot \left(\nabla_{\underline{\theta}} \psi \right)$$

where,

$$\psi(\underline{\theta}) := E_{\underline{\theta}}\left(w(x)\right)$$

and

$$\underbrace{I(\theta)}_{\text{(Fisher information matrix)}} := E_{\underline{\theta}}\left[\frac{\partial}{\partial\theta_i}\log P_\theta(x) \cdot \frac{\partial}{\partial\theta_j}\log P_\theta(x)\right]$$

2) CLRB for iid observations —

$$\underline{X} = (X_1 \ldots X_n) \overset{iid}{\sim} P_\theta \,, \quad \theta \in \Theta \subseteq \mathbb{R}$$

$W(X_1, \ldots X_n)$, CRLB $\Rightarrow$

$$Var_\theta(W) \geq \frac{\left(\frac{d}{d\theta} E_\theta(W)\right)^2}{E_\theta\left(\frac{\partial}{\partial\theta}\sum_{i=1}^{n}\log P_\theta(X_i)\right)^2}$$

$$= \frac{\left(\frac{d}{d\theta} E_\theta W(\underline{X})\right)^2}{n \cdot \underbrace{E_\theta\left[\frac{\partial}{\partial\theta}\log P_\theta(X_1)\right]^2}}$$

Fisher information of single sample.

3) Useful way to calculate Fisher information —

Lemma: if $\{P_\theta(x)\}_{\theta, x}$ satisfy :

$$\frac{d}{d\theta} E_\theta\left[\frac{\partial}{\partial\theta}\log P_\theta(x)\right] = \int \frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}\log P_\theta(x)\right) P_\theta(x)\, dx$$

Then,

$$E_\theta\left(\frac{\partial}{\partial\theta}\log P_\theta(x)\right)^2 = -E_\theta\left[\frac{\partial^2}{\partial\theta^2}\log(P_\theta(x))\right]$$

example CRLB for unbiased estimator of poisson rate ÷

$X_1 \dots X_n \overset{iid}{\sim} \text{Poi}(\lambda), \quad \lambda > 0$

Recall: $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$ and $\frac{1}{(n-1)} \sum_{i=1}^{n} (X_i - \bar{X})^2$

were shown to be unbiased for estimating λ.

$\text{Var}_\lambda(\bar{X}) = \lambda/n$

CRLB: if $W(X_1, \dots X_n)$ is unbiased for estimating λ, then (assuming it satisfies CRLB Conditions)

$$\text{Var}_\lambda W \geq \frac{1}{-n \cdot E_\lambda\left[\frac{\partial^2}{\partial\lambda^2}(\log P_\lambda(x))\right]}$$

$$P_\lambda(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\log_e P_\lambda(x) = \left(-\lambda + x \log_e \lambda - \log_e x!\right)$$

$$\frac{d}{d\lambda} \log_e P_\lambda(x) = \left(-1 + \frac{x}{\lambda}\right)$$

$$\frac{\partial^2}{\partial\lambda^2} \log_e P_\lambda(x) = \left(\frac{-x}{\lambda^2}\right)$$

$$\text{Var}_\lambda(W) \geq \frac{1}{-n \cdot E_\lambda\left[\frac{-x}{\lambda^2}\right]} = \frac{1}{-n\left(\frac{-1}{\lambda^2}\right) E_\lambda[x]}$$

$$= \lambda/n$$

$\text{Var}_\lambda(W) \geq \lambda/n \Rightarrow$ ($\bar{X}$ is a Best unbiased estimator)

example — CRLB for normal parameter estimation →

$X_1 \dots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

if W is an unbiased estimator of μ, then

CLRB $\Rightarrow$ $\mathrm{Var}(W) \geq \dfrac{1}{-n(-1/\sigma^2)} = \sigma^2/n$

implies : $\bar{X}$ is a best unbiased estimator

→ **Unbiased estimation of (μ, σ^2)**

$W : \mathbb{R}^n \longrightarrow \mathbb{R}^2$

if $W_2(X_1 \dots X_n)$ is an unbiased estimator for σ^2, then

CRLB is $\dfrac{1}{n FI_{\sigma^2}(X_1)}$

$\log \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$\theta = \sigma^2$, $-E_\theta\left[\dfrac{\partial^2}{\partial \theta^2} \log P_\theta(x) \right]$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \left\{ -\frac{1}{2}\log(\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2} \right\}$$

$$= \frac{1}{2\sigma^4} - (x-\mu)^2 \times \left(\frac{1}{2} \times \frac{2}{\sigma^6} \right)$$

$$= \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}$$

$$E_\theta\left(\frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6} \right)$$

$$E_{\sigma^2}\left(\frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6} \right) = \frac{1}{2\sigma^4} - \frac{1}{\sigma^4}$$

$$= \frac{-1}{2\sigma^4}$$

Hence, $$\underline{\underline{Var(W_2) \geq \frac{2\sigma^4}{n}}}$$

Recall $$Var\left(\frac{1}{n-1}\sum_i (x_i - \bar{x})^2\right) = \frac{2\sigma^4}{n-1}$$

Que: Is there a estimator for σ^2 (not depending on μ) that meets the CRLB?

No!